The Washington Post

LOUIS PASTEUR AND QUESTIONS OF FRAUD

By **Cristine Russell**

February 23, 1993

BOSTON -- Louis Pasteur's achievements rank him as one of the greatest scientists of all time. But in at least two significant cases, the 19th century French researcher apparently lied about his scientific methods, appropriated an idea from a competitor and conducted human experiments that would be considered unethical then as well as today.

Princeton history professor Gerald L. Geison, a leading Pasteur scholar, unearthed "two examples of what might be called scientific misconduct" in the public trial of a vaccine against deadly anthrax disease in sheep and the vaccination of a young boy against the lethal disease rabies.

Geison concluded that, despite his monumental contributions to medicine, Pasteur would "not have passed muster with Congress" if his scientific methods were scrutinized today.

Princeton history professor Gerald L. Geison, a leading Pasteur scholar, unearthed "two examples of what might be called scientific misconduct" in the public trial of a vaccine against deadly anthrax disease in sheep and the vaccination of a young boy against the lethal disease rabies.

Geison concluded that, despite his monumental contributions to medicine, Pasteur would "not have passed muster with Congress" if his scientific methods were scrutinized today.

"He fully deserves his reputation as one of the greatest scientists who ever lived," said Geison, in a lecture at the recent meeting here of the American Association for the Advancement of Science. But "he was by no means always humble, selfless, ethically superior . . . Quite the opposite."

Geison noted that Pasteur's notebooks revealed that he had regularly "massaged" or manipulated raw data in order to fit his own preconceived ideas. This conforms, he said, with other historical studies that have shown that even among prominent scientists "there are always discrepancies between the private record and published results."

Geison said in an interview that he found Pasteur's "behavior and conduct in general unlikeable through much of his career. He is not a very appealing human being." Geison plans to publish a book on the private science of Pasteur in 1995, the centennial of his death.

Pasteur, who died at age 73, became a French national hero for work ranging from the heat process to kill germs -- now called pasteurization -- to a vaccine for rabies. "After the rabies vaccine, he was Elvis, Madonna and Michael Jackson all rolled into one . . . He was a figure of glory, and he lived to see it," said Geison.

Behind Pasteur's carefully cultivated public image, however, were chinks in his armor that lay hidden in elaborate scientific notebooks and other private documents on his life's work. He carefully instructed his family never to show them to anyone, a request that was honored until the private manuscripts were finally given to the Bibliotheque Nationale in Paris by his last male descendant and became available to scholars in the early 1970s.

Geison said the vast collection included 30 bound volumes of unpublished correspondence, lecture notes and school records and more than 100 laboratory notebooks amounting to perhaps 10,000 pages covering Pasteur's 40-year scientific career. It took a year just to learn to read Pasteur's pinched handwriting, but the Princeton professor eventually found "ethically dubious conduct" in Pasteur's famous anthrax and rabies vaccines.

The anthrax episode began with an article in which Pasteur announced a new vaccine could be produced by exposing the deadly anthrax organism to oxygen to reduce its strength. Then, he said, the organism was suitable for vaccination to protect animals from the infectious disease. He impulsively accepted a public challenge to demonstrate its effectiveness, despite the private consternation of his own colleagues and his own inner doubts that his vaccine was really ready for use.

In May 1881, in the French village of Pouilly-le-Fort, Pasteur launched a "bold public demonstration" of the vaccine in an experiment involving 50 sheep. When it was successful, he returned in June "to bask in the applause" from an appreciative crowd of reporters and dignitaries. "It was a moment of high drama," said Geison.

According to a footnote in one of his lab notebooks, however, Pasteur had used another vaccine approach in which the anthrax was treated with chemicals to weaken it rather than being exposed to oxygen. Using this approach, suggested Geison, Pasteur "pushed aside a rival," Jean-Joseph Henri Toussaint, an obscure veterinarian who developed the chemically treated vaccine first and had visited Pasteur's laboratory to discuss it.

"Pasteur deliberately deceived the public and scientific community about the precise nature of the vaccine he used," said Geison, calling it a "clear case of scientific misconduct . . . He knew full well he was lying."

Later on, Pasteur pursued his own oxygen-attenuated approach and it became the preferred method, so "his hunch paid off," said Geison. Toussaint suffered a nervous breakdown and died.

Another example of questionable conduct concerned the first human use of the rabies vaccine. On July 5, 1885, a 9-year-old Alsatian peasant, Joseph Meister, showed up unannounced at Pasteur's door. He had been bitten 14 times by a dog with classic signs of rabies. While young Meister would not know for several weeks if he were infected, Pasteur decided to administer his new rabies vaccine in hopes of saving the youth's life.

Meister did survive, and three months later Pasteur published a paper reporting that his rabies vaccine had previously been tested on 50 dogs without a single failure before he used it to treat the boy. But Geison discovered through the notebooks that this was, "to put it charitably, a very misleading account."

In fact, Pasteur had extensively tested a vaccine on dogs that used an approach that was exactly the reverse of the one used on Meister. The method he used on the boy involved injections of successively stronger doses of rabies virus. This approach was being tested on laboratory dogs at the time the human experiment was attempted, but Pasteur had no conclusive animal results to show that the technique worked.

"There was no experimental evidence for his published claims about the extent of the safety and efficacy of the vaccine in animals before the human rabies trial," said Geison. "He guessed right," however, paving the way for a flood of other animal-bite victims to benefit from the vaccine. The Pasteur Institute, which went on to worldwide fame as a biomedical research facility, was founded in 1888 as a rabies vaccination center.

While sharply critical of Pasteur in the anthrax episode, Geison is more ambivalent about the rabies case. He notes that the ethical standards of the late 19th century did require extensive animal studies before human experimentation. And Pasteur himself had proclaimed that such proof was needed. A medical colleague of Pasteur's, Emile Roux, refused to participate in the Meister rabies trial on the grounds it was "an unethical form of human experimentation."

On the other hand, Geison notes, when Meister showed up at Pasteur's door, the scientist was "face to face" with a boy who was probably doomed and, in a sense, the rabies experiment was an "act of courage and humanitarianism." With deadly diseases, suggested Geison, ethical standards "ordinarily applied to other cases should be somewhat relaxed."

Pasteur's message for contemporary science, Geison argued, was to puncture the "hopelessly misleading" image of science as "simply objective and unprejudiced," a myth that scientists have perpetuated in order to advance their work and attain a "privileged status."